

PHYS 598: Physics of Living Systems  
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### 3. Gene regulation as a dynamical system

How signal reaching the DNA controls gene expression and cell's reversible decision-making

In this chapter, we will discuss how to model standard gene-regulatory schemes - constitutive gene expression, positive feedback, and negative feedback. We will analyze the deterministic aspects and then study their stochastic behaviors.

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## 1 From sensing the environment to regulating a gene

Now suppose that the cell has detected a molecule outside and relayed the signal inside. What can it do with this signal? One possibility is to change how much of a protein it makes. A kinase can activate a transcription factor, which binds to DNA and changes the rate at which a gene produces mRNA. The cell then uses the mRNA to make the protein.

A change in the production rate does not immediately tell us the protein concentration. The cell also loses proteins through degradation, and growth can dilute their concentration. We therefore need to keep track of both production and loss. The protein that is made can, in turn, change its own production rate or the production rate of another protein. We then have feedback.

The names and functions of these proteins can be very different. Yet the equations that describe their concentrations can have the same form. We will study a few such equations and ask how a cell responds to a signal, how quickly it responds, and under what conditions it can retain a memory after the signal disappears.

## 2 Basics of gene regulation

**Gene expression** is the process of making RNA and protein from a gene (Fig. 1). Genes are particular sequences on DNA. We say that a gene is expressed if the molecules that the gene sequence codes for - RNA and (usually) the protein - are made. More specifically, if a protein is coded by the gene, then the RNA that is made from the gene is called **mRNA (messenger RNA)**. The mRNA in turn contains the sequences that encode the protein, both its content and shape. We say that mRNA is **transcribed** from the gene on the DNA - this process is called **transcription** - and that a protein is **translated** from the mRNA - this process is called **translation**. Specifically, particularly big enzymes, which are called macromolecular machines due to their size (because they are usually formed from multiple proteins), govern transcription and translation. One of these macromolecular machines, called **RNA polymerase** lands on DNA and slides over the sequence that encodes a gene to transcribe mRNA. Specifically, as the RNA polymerase slides over the gene, letter-by-letter (A, T, C, G), it grabs a single base (i.e., a single letter), called a **nucleotide**, that is floating around inside the cell and joins it to the next letter that it grabs (which corresponds to the next letter of the gene sequence). As a result, once the RNA polymerase has finished running through the entire gene sequence, it will have assembled a string of single-bases (i.e., a string of letters that look like one half of a ladder) - this is the mRNA encoded by the gene. Once this transcription of mRNA is complete, another macromolecular machine, called the **ribosome**, lands on the mRNA and slides over it, from start to end, to synthesize a protein that the mRNA encodes (and thus the gene encodes). As the ribosome runs through the mRNA, nucleotide by nucleotide, it grabs onto **amino acids** that are floating inside the cell and then, like the RNA polymerase, attach one amino acid to the next, with each amino acid corresponding to *three* consecutive nucleotides of mRNA - called a **codon**. Once assembled, the string of amino acids folds, like an origami, into a three-dimensional shape that we call a **protein**. How this folding occurs is still not yet fully understood and is an active topic of research in biology. Each protein is a molecular machine that performs a certain function inside the cell. The protein's shape and amino-acid composition determines what it does. For example, the protein can be an enzyme for certain processes or it can form part of the cellular structure (e.g., cell wall) like a brick in a building. Despite the wide variety of proteins that exist inside a cell and thus the genes that code the proteins may be - the human cell roughly has 20,000 genes that code for a protein - only a few types of differential equations dictate the concentrations of proteins and of mRNAs inside cells. The specific form of the equation is determined not by the name of the proteins or the genes, but rather by which type of **gene-regulation scheme** controls those genes. In this section, we derive and analyze the

equations for some widely-occurring types of gene-regulations.

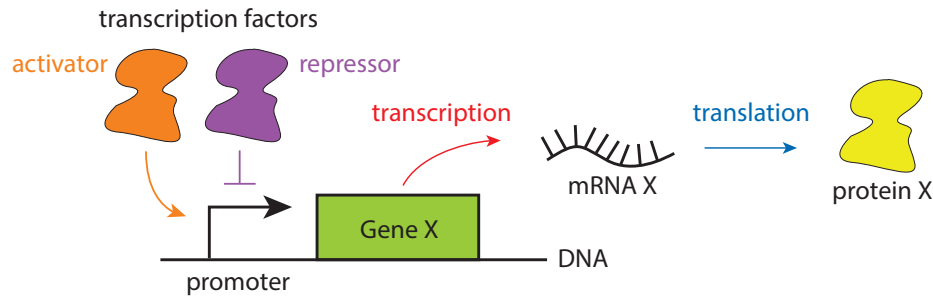


Figure 1: **Central dogma of molecular biology:** A gene (green box) is a sequence of DNA (black horizontal line) and a promoter is the DNA sequence in front of the gene. The promoter is a "molecular dial" that controls how much mRNA (and thus protein) is transcribed from the gene. During translation of a protein from an mRNA, the mRNA is *not* destroyed. Thus, the cell can translate multiple copies of a protein from a single mRNA molecule. Broadly speaking, two types of proteins (transcription factors), called activators and repressors, regulate the transcription of a gene. An activator binds to the promoter to increase the transcription rate (more mRNA produced per unit time) while a repressor binds to the promoter to decrease the transcription rate (less mRNA produced per unit time). A regular arrow (orange arrow) indicates an activation of a gene (the action of the activator) while a blunt-ended arrow (purple arrow) indicates a repression of a gene (the action of the repressor).

**Gene regulation** is a dynamic process in which the cell modulates its gene expression - the cell dynamically modifies the amounts of mRNA and protein according to some rule. As we will see, different rules can regulate gene expression. We are usually interested in the concentration of the gene's final product, the protein, as a function of time. Most regulations of a single gene lead to an equation of the following type:

$$\frac{dx}{dt} = F - G \quad (1)$$

where  $x$  is the concentration of a protein-X (made from gene X),  $F$  and  $G$  are non-negative-valued functions that can depend on  $x$  and the concentrations of proteins translated from other genes.  $F$  is a "**production-rate function**" that describes how much protein is made per unit time and  $G$  is a "**Degradation-rate function**" that describes how much protein is degraded per unit time. If we are thus interested in genes X, Y, and Z, for example, we would write down an equation of the following form:

$$\frac{dx}{dt} = F_1 - G_1 \quad (2a)$$

$$\frac{dy}{dt} = F_2 - G_2 \quad (2b)$$

$$\frac{dz}{dt} = F_3 - G_3 \quad (2c)$$

where  $F_1$ ,  $F_2$ , and  $F_3$  are production-rate functions for proteins X, Y, and Z, respectively, and  $G_1$ ,  $G_2$ , and  $G_3$  are degradation-rate functions for proteins X, Y, and Z, respectively. Note that Eqs. 2a - 2c can be coupled amongst each other. Moreover, note that if we have  $N$  genes, then we would have  $N$  equations that may be potentially coupled to each other. This already tells us why analyzing more than a few genes is difficult. As we will see, even for a regulation scheme of a single gene, the equation (Eq. 1) is complicated enough that we cannot analytically solve it. Yet, we now turn to analyzing equations of this type for some commonly occurring gene regulatory schemes.

### 3 Gene-regulation scheme 1: Constitute gene expression

At its simplest, gene expression can be captured by the following process:



where  $X$  is the name of a gene,  $DNA_X$  is the DNA molecule that contains the gene sequence,  $mRNA_X$  is the mRNA transcribed from gene  $X$ , and  $protein_X$  is the protein translated from  $mRNA_X$ . Here, gene- $X$  produces one type of mRNA and one type of protein, each at some constant rate. Let  $k_m$  be the transcription rate (i.e., rate associated with the chemical reaction in Eq. 3a) and  $k_p$  be the translation rate (i.e., rate associated with the chemical reaction in Eq. 3b). Note that DNA and the RNA polymerase are not consumed in this process shown in Eq. 3a. Thinking of RNA polymerase as an enzyme, this makes sense - like any enzyme, it is not destroyed in any reactions. Note that DNA also is not destroyed during transcription. If it did, it would be like burning a book (DNA) after copying all the words in the book (sequence of letters), which wouldn't make for a good library (cell)! Cells indeed do the sensible thing and keeps the DNA intact. Eq. 3a is thus like the enzyme-substrate kinetics except that here, we have two enzymes - RNA polymerase and  $DNA_X$  - that yield a product,  $mRNA_X$ , without any substrates. Note also, in Eq. 3b, that the mRNA and the ribosome are not consumed during the translation of  $protein_X$ . Unlike the RNA polymerase, DNA, and the ribosome, the  $mRNA_X$  and the  $protein_X$  do degrade. We represent these degradations as



where  $\emptyset$  represents "nothingness" (i.e., symbol for null). Let  $\gamma_m$  be the constant degradation rate of the  $mRNA_X$  (reaction in Eq. 4a) and  $\gamma_p$  be the constant degradation rate of the  $protein_X$  (reaction in Eq. 4b). In addition to not having any degradations of the Ribosome, DNA, and the RNA polymerase, we will assume that their concentrations do not change over time. That is, we assume that their concentrations inside the cell remain constant throughout the life of the cell. This is a good assumption for many situations. One place where this assumption can fail is during the cell cycle in which the cell replicates all of its DNA. This assumption also fails during physical replication of the cell itself (one cell becoming two cells) if the molecule number does not keep up with the increase in the cell's volume. With these caveats in mind, the above four chemical kinetics equations - Eqs. 3a- 4b - yield the following equations:

$$\frac{dm}{dt} = k_m[DNA_X][\text{RNA polymerase}] - \gamma_m m \quad (5a)$$

$$\frac{dp}{dt} = k_p m[\text{Ribosome}] - \gamma_p p \quad (5b)$$

where  $m = [mRNA_X]$  and  $p = [protein_X]$ . Since  $[DNA_X]$ ,  $[\text{RNA polymerase}]$  and  $[\text{Ribosome}]$  do not change over time, we can simplify Eqs. 5a and 5b by defining new constants:  $r_m = k_m[DNA_X][\text{RNA polymerase}]$  and  $r_p = k_p[\text{Ribosome}]$ . Then Eqs. 5a and 5b become

$$\frac{dm}{dt} = r_m - \gamma_m m \quad (6a)$$

$$\frac{dp}{dt} = r_p m - \gamma_p p \quad (6b)$$

Above two are coupled differential equations. They are coupled because the protein concentration  $p$  depends on the mRNA concentration  $m$ . We would like to determine  $m$  and  $p$  as functions of time (i.e., determine  $m(t)$  and  $p(t)$ ). We can solve for  $m(t)$  by hand. The tricky part is finding  $p(t)$  because it depends on  $m(t)$ , which changes over time. We can solve for  $p(t)$  exactly by hand by using an integrating factor. But its solution contains more terms. So we will first simplify the equations by making a biologically reasonable assumption. Note that we can solve above equations by hand if  $p(t)$  does not depend on the  $m$ , or if  $m$  does not change over time. Indeed our saviour is that for typical genes, the changes to their mRNA concentrations occur much faster than the changes to the concentrations of the proteins that these mRNAs encode. That is, we can assume that the mRNA concentration quickly reaches a steady-state value before the protein concentration changes appreciably. This is called a **quasi steady-state assumption**. Then Eqs. 6a and 6b become

$$\frac{dm}{dt} = r_m - \gamma_m m \quad (7a)$$

$$\frac{dp}{dt} = r_p m_{eq} - \gamma_p p, \quad (7b)$$

where  $m_{eq}$  is the constant, equilibrium-state concentration of the mRNA. Now note that Eqs. 7a and 7b have mathematically the same form but with different variables. Let us solve for  $m_{eq}$ . By definition of being a steady state, we must have  $dm/dt = 0$  when  $m = m_{eq}$ . Then Eq. 7a gives us

$$m_{eq} = \frac{r_m}{\gamma_m}, \quad (8)$$

and thus Eq. 7b becomes

$$\frac{dp}{dt} = \frac{r_p r_m}{\gamma_m} - \gamma_p p, \quad (9)$$

Eq. 8 makes sense: The higher the production rate of the mRNA, the more of the mRNA there is inside a cell. The higher the degradation rate of the mRNA, the less of the mRNA there is inside the cell. Fully solving for  $dm/dt$  tells us how the mRNA concentration reaches this steady-state over time. To solve equations like  $dm/dt$  and  $dp/dt$  - the later now has the same form as  $dm/dt$  - we use the following two-step procedure. First, taking Eq. 7a, we solve for the simpler case in which there is no production. In this case, Eq. 7a becomes

$$\frac{dm_1}{dt} = -\gamma_m m_1, \quad (10)$$

where we relabelled  $m$  as  $m_1$  (just a different variable name so that we do not confuse ourselves later, as we will see).

Solving Eq. 10 yields

$$m_1(t) = A e^{-\gamma_m t} \quad (11)$$

where  $A$  is a constant whose value we will later determine. Next, we look for a particular solution  $m_2$  for the original equation, Eq. 7a:

$$\frac{dm_2}{dt} = r_m - \gamma_m m_2, \quad (12)$$

where we have relabeled  $m$  as  $m_2$  (again, just a different variable name so that we do not confuse ourselves later, as we will see). Here, we would succeed in finding a solution that satisfies Eq. 12. To do so, we note that if  $m_2 = \text{constant}$  is a solution to Eq. 12). In particular, the following is a solution to Eq. 12):

$$m_2 = \frac{r_m}{\gamma_m} \quad (13)$$

In fact, this  $m_2$  is the steady state value that we found earlier. So far, we found  $m_1$  and  $m_2$ .  $m_1$  is the solution to the simpler equation, Eq. 10, and  $m_2$  is the solution to the original equation, Eq. 12. If we add  $dm_1/dt$  to  $dm_2/dt$ , note that we obtain

$$\frac{d(m_1 + m_2)}{dt} = r_m - \gamma_m(m_1 + m_2), \quad (14)$$

Thus  $m_1 + m_2$  is also a solution to the original equation. To see this, note that if we relabel  $m_1 + m_2$  as  $m$ , then above equation becomes

$$\frac{dm}{dt} = r_m - \gamma_m m, \quad (15)$$

which gives us back the original equation that we wanted to solve, Eq. 7a.  $m(t)$  is the most **general solution** to the original equation, Eq. 7a:

$$m(t) = m_1(t) + m_2(t) \quad (16a)$$

$$= \frac{r_m}{\gamma_m} + Ae^{-\gamma_m t} \quad (16b)$$

$m(t)$  contains an undefined constant,  $A$ , which is related to the concentration of mRNA at  $t = 0$ :

$$m(0) = \frac{r_m}{\gamma_m} + A \quad (17)$$

Under the quasi-steady state assumption,  $dp/dt$  (Eq. 9) takes the same form as  $dm/dt$  (Eq. 15). Thus we can use the same two-step procedure that gave us the  $m(t)$ . Or more simply, we just relabel the variable names accordingly in Eq. 16b to obtain the most general solution for  $p(t)$ :

$$p(t) = \frac{r_p r_m}{\gamma_m \gamma_p} + Be^{-\gamma_p t}, \quad (18)$$

where  $B$  is a constant that is determined by the initial condition on  $p$  (i.e.,  $p(0)$ ). Thus, like Eq. 17, we have

$$p(0) = \frac{r_p r_m}{\gamma_m \gamma_p} + B \quad (19)$$

Now that we have deduced  $m(t)$  and  $p(t)$ , let's look at the case in which there are initially no proteins or mRNAs (i.e.,  $p(0) = 0$  and  $m(0) = 0$ ). Then we can solve for  $A$  through Eq. 17 and  $B$  through Eq. 19 to convert Eqs. 16b and 18 into

$$m(t) = \frac{r_m}{\gamma_m} \left( 1 - e^{-\gamma_m t} \right) \quad (20a)$$

$$p(t) \approx \frac{r_p r_m}{\gamma_m \gamma_p} \left( 1 - e^{-\gamma_p t} \right) \quad (20b)$$

Note that the equilibrium-state of the protein concentration, which we get by setting  $dp/dt = 0$ , is  $\frac{r_p r_m}{\gamma_m \gamma_p}$ . This intuitively makes sense (run through the argument yourself as we did above for  $m_{eq}$  to convince yourself). Curiously, note that the time to reach the equilibrium-state value is determined only by the degradation rates,  $\gamma_p$  and  $\gamma_m$ , because only the degradation rates appear inside the exponentials.

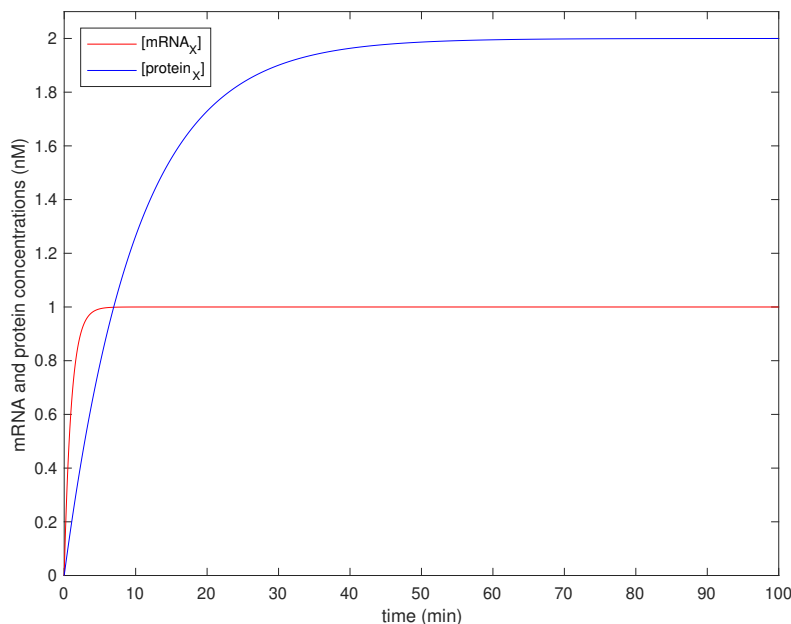


Figure 2: **mRNA and protein concentrations for a constitutive gene expression (Plot of Eqs. 20a and 20b)**: Red curve is  $[mRNA_x]$  as a function of time (i.e.,  $m(t)$  in Eq. 20a) and blue curve is  $[protein_x]$  as a function of time (i.e.,  $p(t)$  in Eq. 20b). Values for the parameters are:  $r_m = 1$  nM/min,  $r_p = 0.2$  min $^{-1}$ ,  $\gamma_m = 1$  min $^{-1}$ , and  $\gamma_p = 0.1$  min $^{-1}$ . Here  $nM$  means nanomolar (nano =  $10^{-9}$ ).

Plotting  $m(t)$  and our approximate  $p(t)$  as functions of time (Fig. 2), we see that both curves have the same shape. This makes sense since Eqs. 20a and 20b have the same mathematical form. The mRNA concentration rises faster because we chose  $\gamma_m$  to be ten times larger than  $\gamma_p$ . The mRNA solution is exact. But the protein solution assumes that the mRNA concentration has already reached its steady-state value. What do we miss by making this assumption? We can answer this by solving the original equations exactly.

**Exact solution without the quasi steady-state approximation:** Let  $m_0 = m(0)$  and  $p_0 = p(0)$  be the initial concentrations. We will write the steady-state concentrations as

$$m_{eq} = \frac{r_m}{\gamma_m}, \quad p_{eq} = \frac{r_p m_{eq}}{\gamma_p} = \frac{r_p r_m}{\gamma_m \gamma_p}. \quad (21)$$



Here “steady state” means that production balances loss. It does not mean that the cell is in thermodynamic equilibrium. From Eq. 16b, we already know the exact mRNA concentration:

$$m(t) = m_{eq} + (m_0 - m_{eq})e^{-\gamma_m t}. \quad (22)$$

We can put this expression into the original protein equation, Eq. 6b. To solve that equation, first put the protein-loss term on the left and multiply both sides by  $e^{\gamma_p t}$ :

$$\begin{aligned} \frac{dp}{dt} + \gamma_p p &= r_p m(t), \\ e^{\gamma_p t} \frac{dp}{dt} + \gamma_p e^{\gamma_p t} p &= r_p e^{\gamma_p t} m(t), \\ \frac{d}{dt} (e^{\gamma_p t} p(t)) &= r_p e^{\gamma_p t} m(t). \end{aligned} \quad (23)$$

The product rule gives us the last line. This is why we chose the **integrating factor**  $e^{\gamma_p t}$ . We can now integrate both sides from 0 to  $t$ :

$$p(t) = e^{-\gamma_p t} \left[ p_0 + r_p \int_0^t e^{\gamma_p s} m(s) ds \right]. \quad (24)$$

The variable  $s$  just labels time inside the integral. Substituting Eq. 22, we have two integrals to calculate:

$$\int_0^t e^{\gamma_p s} m(s) ds = m_{eq} \frac{e^{\gamma_p t} - 1}{\gamma_p} + (m_0 - m_{eq}) \frac{e^{(\gamma_p - \gamma_m)t} - 1}{\gamma_p - \gamma_m}, \quad (25)$$

where we have assumed  $\gamma_m \neq \gamma_p$ . Putting this into Eq. 24 gives the exact solution:

$$p(t) = p_{eq} + (p_0 - p_{eq})e^{-\gamma_p t} + \frac{r_p(m_0 - m_{eq})}{\gamma_m - \gamma_p} (e^{-\gamma_p t} - e^{-\gamma_m t}). \quad (26)$$

The first two terms are the solution we obtained by holding the mRNA concentration at  $m_{eq}$ . The last term accounts for the time that the mRNA concentration takes to approach  $m_{eq}$ . If  $m_0 = m_{eq}$ , this term vanishes. In that particular case, the simpler protein solution is exact, even without a separation between  $\gamma_m$  and  $\gamma_p$ .

For the case we plotted,  $m_0 = p_0 = 0$ , the exact solution becomes

$$p(t) = p_{eq} \left[ 1 - \frac{\gamma_m e^{-\gamma_p t} - \gamma_p e^{-\gamma_m t}}{\gamma_m - \gamma_p} \right]. \quad (27)$$

Unlike Eq. 20b, this expression contains both decay rates. Protein production has to wait for mRNA to build up. This does not mean that translation waits until the mRNA reaches its steady state. Translation starts as soon as mRNA is available.

What if  $\gamma_m = \gamma_p = \gamma$ ? We cannot substitute directly into Eq. 25 because its second term then has the form 0/0. Instead, the second integrand is just  $m_0 - m_{eq}$ , so its integral is  $(m_0 - m_{eq})t$ . Thus

$$p(t) = p_{eq} + (p_0 - p_{eq})e^{-\gamma t} + r_p(m_0 - m_{eq})te^{-\gamma t}. \quad (28)$$

For  $m_0 = p_0 = 0$ , this reduces to

$$p(t) = p_{eq} [1 - (1 + \gamma t)e^{-\gamma t}], \quad p_{eq} = \frac{r_p r_m}{\gamma^2}. \quad (29)$$

There is nothing singular about the dynamics when the two rates are equal. Only the way we wrote Eq. 26 needs to change.

**When does the approximation miss the dynamics?** First, consider the initial response when  $m_0 = p_0 = 0$ . At  $t = 0$ , the original protein equation gives  $dp/dt = 0$  because there is no mRNA yet. At very short times,  $m(t) \approx r_m t$ . Thus, to leading order,

$$\frac{dp}{dt} \approx r_p r_m t, \quad p(t) \approx \int_0^t r_p r_m s ds = \frac{r_p r_m}{2} t^2. \quad (30)$$

The exact protein concentration initially grows quadratically in time. But our quasi steady-state solution, Eq. 20b, initially grows linearly:

$$p(t) \approx p_{eq} \gamma_p t = \frac{r_p r_m}{\gamma_m} t. \quad (31)$$

The approximation supplies the cell with its full steady-state mRNA concentration from the start. It therefore misses this initial delay.

Second, the approximation can miss much more than the initial delay if  $\gamma_m$  is similar to or smaller than  $\gamma_p$ . The mRNA then does not relax much faster than the protein, and we cannot ignore its changing concentration. We should use Eq. 27, or Eq. 29 for equal rates. In particular, slow mRNA dynamics can set the time over which protein builds up.

For  $\gamma_m \gg \gamma_p$ , the mRNA relaxes on the short time scale  $1/\gamma_m$ . For zero initial concentrations, the simpler protein solution describes the slower response well after this initial period. The difference between the approximate and exact protein solutions is

$$p_{\text{approx}}(t) - p_{\text{exact}}(t) = p_{eq} \frac{\gamma_p}{\gamma_m - \gamma_p} (e^{-\gamma_p t} - e^{-\gamma_m t}). \quad (32)$$

Here  $p_{\text{approx}}$  is Eq. 20b and  $p_{\text{exact}}$  is Eq. 27. The difference is small compared to  $p_{eq}$  when  $\gamma_p/\gamma_m$  is small. But it can still be large compared to the very small protein concentration at early times. So the approximation can describe the overall rise well while getting the initial response wrong.

For other initial conditions, we should also check the last term in Eq. 26. A cell that starts with a very large excess of mRNA can produce a substantial burst of protein while that mRNA decays. A separation of decay rates alone does not let us discard this burst.

We have assumed constant production rates in these calculations. If an external signal changes  $r_m$  with time, we need one more check. The mRNA must also relax faster than the signal changes before we can replace  $m(t)$  by the instantaneous value  $r_m(t)/\gamma_m$ . A short pulse can end before the mRNA has caught up, even when  $\gamma_m \gg \gamma_p$ .

**Validity of the quasi steady-state approximation:** Can we really use the quasi steady-state approximation in real cells? To answer this question, we need to look up "typical" values of degradation and production rates of mRNA and proteins. To look up the typical numbers such as the typical time for a cell to replicate, the number of bases in the human genome, and other useful numbers, you can look at [www.BioNumbers.org](http://www.BioNumbers.org). Alternatively, you can see: "Key numbers in biology" (Moran, Phillips, and Milo, *Cell* (2010)). The typical life-time of mRNA is indeed shorter than the typical life-time of protein in many cases (see Table 1). By "life time", we mean "half life". Half life is the time taken for the concentration of a molecule to decrease to half of its current value.

| Organism                                          | half-life of typical mRNA |
|---------------------------------------------------|---------------------------|
| E. coli                                           | 5 minutes                 |
| Budding yeast ( <i>Saccharomyces cerevisiae</i> ) | 20 minutes                |
| Human cell                                        | 600 minutes               |

Table 1: From: Moran, Phillips, and Milo, *Cell* (2010)

Now, what about half-life of proteins? Many proteins, including the fluorescent proteins such as the GFP (Green Fluorescent Protein) that researchers often use to measure gene expression in a cell, are very stable. This means that if the cell were not to divide and yet live happily, the GFP does not degrade because there is no active mechanism in a cell that would degrade it. The concentration can decrease to half its current value as the cell grows to twice its volume without making more of the protein. In this process, the mother (original) cell divides into two cells (i.e., the mother gives rise to a daughter cell), and in this process the mother cell must give some of its proteins to the daughter cell. For most biomolecules, the mother cell has no way of counting or regulating many proteins she gives to her daughter. Our differential equations describe the average concentrations inside a cell. On average, we expect that the mother to get the half of the proteins and the daughter to inherit the remaining half of the proteins. At a symmetric division, both molecule number and cell volume halve, so the concentration does not halve at division. It is the growth in volume between divisions that dilutes the protein. Thus the effective degradation rate  $\gamma_p$  is set by the doubling time of the cell. We can calculate  $\gamma_p$  in terms of the doubling time  $\tau$ . The doubling time is the average time taken for a cell to grow in size (approximately twice its initial size) and then divide into two, thus going from one cell to two cells. Let  $N(t)$  be the integer number of protein X inside a cell at time  $t$  and  $V(t)$  be the volume of the cell at time  $t$ .  $t = 0$  is the beginning of a cell cycle (for the daughter cell, its cell cycle is just after it is born). Then the concentration inside the cell is  $p(t) = N(t)/V(t)$ . Thus, we have

$$\frac{dp}{dt} = \frac{d(N/V)}{dt} \quad (33a)$$

$$= \frac{1}{V} \frac{dN}{dt} - \frac{N}{V} \frac{1}{V} \frac{dV}{dt} \quad (33b)$$

Confining ourselves to the inside of just one cell, we see that the volume changes over time as

$$V(t) = V_0 2^{t/\tau} \quad (34a)$$

$$= V_0 e^{\mu t} \quad (34b)$$

where  $V_0$  is the volume of the cell at  $t = 0$  and  $\mu = \ln(2)/\tau$  is the cell's growth rate. Then computing  $dV/dt$ , we find that  $dp/dt$  becomes

$$\frac{dp}{dt} = \frac{1}{V} \frac{dN}{dt} - \mu p \quad (35)$$

Assuming that a constant translation rate  $k_N$  for the absolute (integer) number of proteins (i.e.,  $dN(t)/dt = k_N$ ), above equation becomes

$$dp/dt = k_N/V - \mu p \quad (36)$$

Eq. 36 is identical to Eq. 9 if the production rate in terms of concentration  $k_N/V$  were constant. But note that since the cell volume can double within the doubling time (i.e., cell cycle time), the production rate is not constant over time. The production rate  $k_N/V$  can decrease by as much half. But we ignore this and say that the production rate is constant (i.e., replace the  $k_N/V$  by a constant) and set  $\gamma_p = \mu$ . In this sense, we have just derived Eq. 9 in a different way. The important point here is that for very stable proteins, if you know the doubling time of the cell, then you know the dilution rate (i.e., degradation rate) of the highly stable protein in the equation for  $dp/dt$ . The typical values for cell division times are shown in Table 2.

| Organism                                          | Typical cell-division (doubling) time |
|---------------------------------------------------|---------------------------------------|
| E. coli                                           | 30 minutes                            |
| Budding yeast ( <i>Saccharomyces cerevisiae</i> ) | 90 minutes                            |
| Human cell                                        | 3000 minutes                          |

Table 2: Typical doubling times of cells

According to Eqs. 20a and 20b, within the quasi steady-state approximation, the mRNA and protein relaxation times are  $1/\gamma_m$  and  $1/\gamma_p$ , respectively. Thus the ratio of  $1/\gamma_m$  to  $1/\gamma_p$  tells us how much faster mRNA concentration relaxes compared to the protein concentration. Using Tables 1 and 2, we can compute these values (see Table 3). For exponential relaxation, the half-life is  $\ln(2)/\gamma$ , whereas the relaxation time is  $1/\gamma$ . Thus, the values in the tables give relaxation times of about 7 minutes for the mRNA and 43 minutes for a stable protein in E. coli. These are times to complete about 63% of the approach to steady state, not times at which the steady state is reached exactly. This separation of times supports the quasi steady-state approximation for the slower protein response in these "typical" mRNA and proteins in E. coli. We should still check the initial response and the time scale of any changing input, as discussed above.

| Organism                                          | $\gamma_p/\gamma_m$ |
|---------------------------------------------------|---------------------|
| E. coli                                           | 0.17                |
| Budding yeast ( <i>Saccharomyces cerevisiae</i> ) | 0.22                |
| Human cell                                        | 0.20                |

Table 3: If  $\gamma_p/\gamma_m \ll 1$ , the quasi steady-state approximation is valid.

**Interpretation of the constitutive gene-expression scheme:** At the beginning of this section, we introduced constitutive gene expression as a gene-regulation scheme in which the production rate of mRNA (transcription rate per gene) and protein (translation rate per mRNA) are constant. Yet we found that both the protein and the mRNA concentrations inside a cell reach equilibrium-state values, whereby they do not increase or decrease any further over time. This occurs because the mRNA and the protein degrade over time. From Eq. 6a, we see that as the mRNA concentration increases, the mRNA-degradation rate,  $\gamma_m m$ , also increases while the mRNA-production rate,  $r_m$ , remains constant (this is what the "constitutive" in constitutive gene-expression means). Thus when  $m = 0$  initially, the production rate is higher than the degradation rate, which is initially zero. Thus the mRNA concentration starts to increase. But as this happens, the degradation rate gradually increases from zero to a higher value, eventually equalling the production rate  $r_m$ . When this happens, the mRNA concentration reaches an equilibrium value  $m_{eq}$  (Eq. 8). Looking at Eq. 7b, the same argument works to explain why the protein concentration reaches an equilibrium-state value.

## 4 Gene-regulation scheme II: Autoregulatory positive feedback

We now consider a **positive feedback loop** in which a gene produces a protein that promotes the production of itself (Fig. 3). Here, a gene X produces a protein X that is an **activator** of its own production. This means that protein X binds to the **promoter** of gene X, which increases the transcription rate of X, and in turn, the production rate of protein X. A **promoter** is a sequence of DNA that is just in front of the gene. It controls how well RNA polymerase can bind to that region and then run through the gene - it controls the transcription rate of the *mRNA* from that gene. The regulation scheme we are describing here is called an **autoregulatory positive feedback**. Using Eq. 1, the basic equation that describes

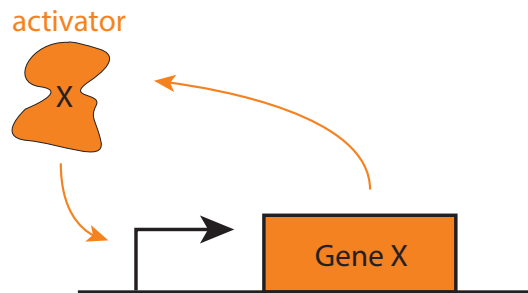


Figure 3: **Auto-regulatory positive feedback loop:** A gene  $X$  encodes a protein that is an activator for the transcription of itself. Thus, the more  $X$  there is inside the cell, the higher the production rate of  $X$ .

the protein concentration, denoted  $p$ , is

$$\frac{dp}{dt} = f(m, p) - g(p) \quad (37)$$

We make one simplification: The production function  $f$  depends only on  $p$ , not  $m$ . So we have

$$\frac{dp}{dt} = f(p) - g(p) \quad (38)$$

We assume that the same type of protein degradation occurs, so  $g(p) = \gamma p$ , where  $\gamma$  is a positive constant that signifies the degradation strength. The production-rate function  $f(p)$  for a positive feedback loop typically takes the following form:

$$f(p) = \frac{Vp^n}{K^n + p^n}, \quad (39)$$

where  $V$ ,  $K$ , and  $n$  are positive constants.  $f(p)$  is a phenomenological equation - it is not derived from anything. It simply has the property that  $f$  increases as  $p$  increases, and that it is non-linear in  $p$ . This non-linearity in  $f$  as a function of  $p$  allows for rich behaviours as we will see. The features that  $f$  embodies makes it widely used for modeling positive feedback (and it fits data too in many instances!). For instance, note that when  $p$  approaches infinity,  $f$  approaches  $V$ . So  $V$  is the maximal production rate. When  $p = K$ ,  $f$  is  $V/2$ . We thus call  $K$  the **half-saturation constant**, because it is the concentration of protein  $p$  at which the production rate is half of the maximum possible value (i.e., half of the saturated rate). The  $n$  is called a **Hill coefficient** and is typically between 1 and 2 for most genes. A Hill coefficient of 2 is, in fact, quite rare in cells and is considered to be a very high number. The Hill coefficient characterizes how **switch-like** the positive feedback is, with a higher value being more switch-like and a value closer to one being less switch-like (see Fig. 4). In Fig. 4, we see that indeed, when the Hill coefficient is incredibly high ( $n = 10$ ), the production rate  $f(p)$  looks like a step-function (thus the name "switch-like"). Specifically, for  $n = 10$ , when  $p$  is smaller than  $K$ ,  $f(p)$  is nearly zero (no production of protein) whereas when  $p$  is larger than  $K$ , then  $f(p)$  approximately equals the maximum possible production rate,  $V$ . Taking a close-up view of the production rate  $f(p)$  at low protein concentrations (i.e., low values of  $p$ ), we see that indeed as the Hill coefficient increases, the closer  $f(p)$  approaches zero when  $p$  is smaller than  $K$ . (see Fig. 5). When  $n = 1$ ,  $f(p)$  is exactly the same as the production rate in the well-known Michaelis-Menten reaction scheme.

Putting together, Eq. 38 is then

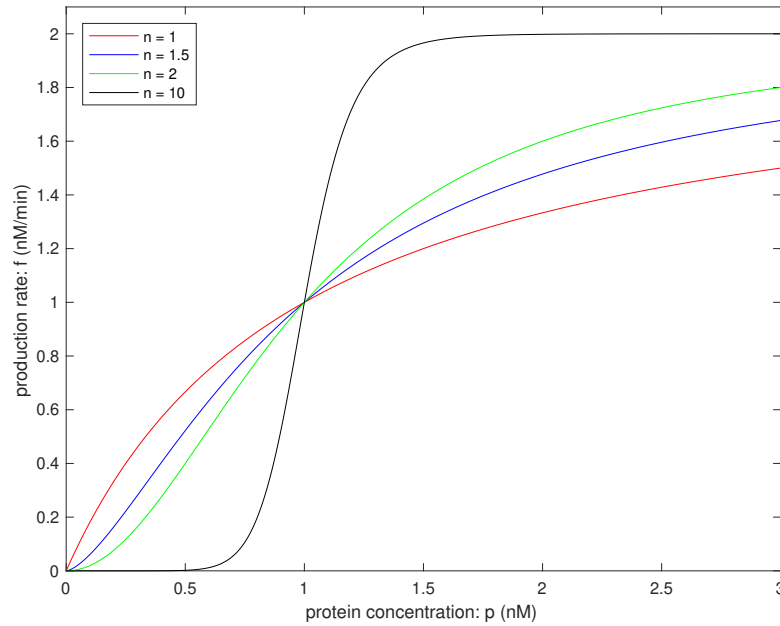


Figure 4: **Production rate for autoregulatory positive feedback (Plot of Eq. 39):** Plot of the production-rate function  $f(p)$  given in Eq. 39. Different colored curves denote different values of the Hill coefficient  $n$ : Red ( $n=1$ ), blue ( $n=1.5$ ), green ( $n=2$ ) and black ( $n=10$ ). For all curves,  $V=2$  nM/min,  $K=1$  nM.

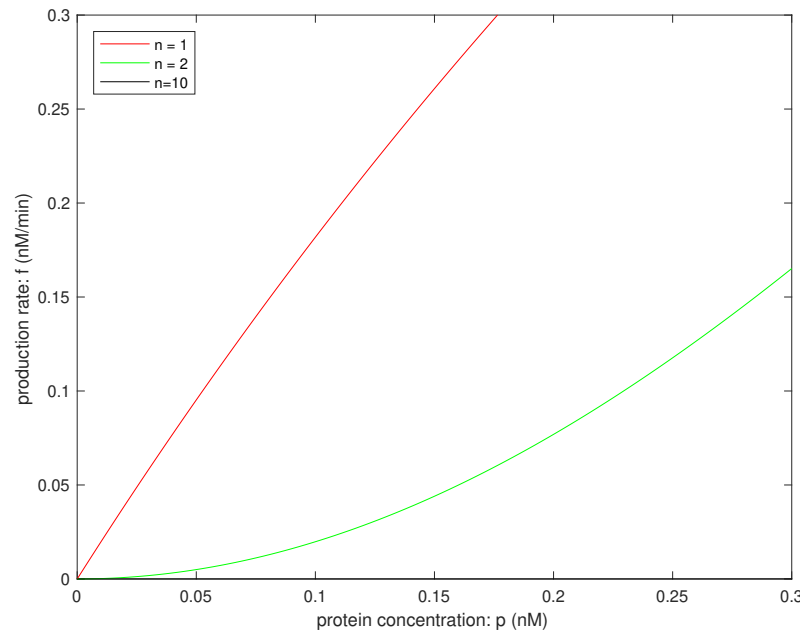


Figure 5: **A close-up view, at low protein concentrations, of the production rate for autoregulatory positive feedback (Plot of Eq. 39):** Plot of the production-rate function  $f(p)$  given in Eq. 39 at low values of  $p$  (i.e., low protein concentrations). Different colored curves denote different values of the Hill coefficient  $n$ : Red ( $n=1$ ), green ( $n=2$ ) and black ( $n=10$ ). For all curves,  $V=2$  nM/min,  $K=1$  nM. The curve for  $n=10$  is a nearly flat line along the horizontal axis (near  $p=0$ ).

$$\frac{dp}{dt} = \frac{Vp^n}{K^n + p^n} - \gamma p \quad (40)$$

There are four constants in Eq. 40 ( $V$ ,  $K$ ,  $n$ , and  $\gamma$ ). Let's reduce this number by making some simplifications. We can do this by choosing to measure the protein concentration  $p$  and time  $t$  in units other than nanomolar and minutes. Instead, we can measure them in multiples of the parameters that we are trying to eliminate. To see this, we first divide both sides of Eq. 40 by  $K$  to obtain

$$\frac{1}{K} \frac{dp}{dt} = \frac{1}{K} \frac{Vp^n}{K^n + p^n} - \gamma(p/K) \quad (41a)$$

$$\Rightarrow \frac{d(p/K)}{dt} = \frac{1}{K} \frac{V(p/K)^n}{1 + (p/K)^n} - \gamma(p/K) \quad (41b)$$

Now we see that all instances of  $p$  now appear as  $p/K$ . This tells us that we can measure  $p$  in multiples of  $K$ . In other words, it is  $p/K$  that is important, not the absolute value of  $p$ , for the dynamics of the protein concentration. We define a new variable,  $\hat{p} = p/K$ . Note that  $\hat{p}$  is a **unitless** variable: It does not have any units because both  $p$  and  $K$  have the same units, the unit of concentration (e.g., nM). But from Eq. 41b, it is not yet clear that we have eliminated the parameter  $K$  since it still appears in Eq. 41b even if we replace all appearances of  $p/K$  with the new variable  $\hat{p}$ . Eliminating the  $K$  from our equation was the whole motivation in the first place! Hold your breath, we are not done yet. Remember, we also wanted to measure the time  $t$  in a different unit. Motivated by the fact that we almost eliminated  $K$  by making  $p$  unitless, we attempt to do the same for  $t$  - making  $t$  unitless as well. To do so, note that  $\gamma$  has units of 1/time. So  $\gamma t$  is unitless. The only appearance of  $t$  in Eq. 41b is in the  $dt$  on the left side of the equation. Noting that  $\gamma dt = d(\gamma t)$  since  $\gamma$  is a constant, we divide both sides of Eq. 41b by  $\gamma$  to obtain

$$\frac{d\hat{p}}{d(\gamma t)} = \frac{1}{\gamma K} \frac{V\hat{p}^n}{1 + \hat{p}^n} - \hat{p} \quad (42)$$

As we did through introducing  $\hat{p}$ , we introduce a new variable  $\hat{t} = \gamma t$ , which is unitless time, to rewrite Eq. 42 as

$$\frac{d\hat{p}}{d(\hat{t})} = \frac{1}{\gamma K} \frac{V\hat{p}^n}{1 + \hat{p}^n} - \hat{p} \quad (43)$$

Finally, we introduce our final new variable,  $\hat{V} = V/(\gamma K)$ .  $\hat{V}$  is unitless as well because  $V$  has unit of concentration/time,  $\gamma$  has unit of 1/time,  $K$  has unit of concentration, and thus all the units cancel. We can rewrite Eq. 43 as

$$\frac{d\hat{p}}{d(\hat{t})} = \frac{\hat{V}\hat{p}^n}{1 + \hat{p}^n} - \hat{p} \quad (44)$$

Note that above equation, other than  $\hat{p}$  and  $\hat{t}$ , there are two constants:  $\hat{V}$ , and  $n$ . Thus we reduced the number of constants from four in the original equation (Eq. 40) to just two in the simplified equation (Eq. 44). Eq. 44 also contains only unitless parameters:  $\hat{V}$ ,  $n$ ,  $\hat{p}$ , and  $\hat{t}$ . The simplification required us to measure the protein concentration and time in relative units:  $\hat{p}$  is the protein concentration in units of  $K$  (so  $\hat{p} = 1$  means that  $p = K$ ) and  $\hat{t}$  is the time measured in units of  $1/\gamma$  (so  $\hat{t} = 1$  means that  $t = 1/\gamma$ ).  $\hat{V}$  describes the maximal production rate of the protein in a unitless form.



Let us now analyze Eq. 44. We cannot solve this equation with just a pen and a paper to obtain the protein concentration  $\hat{p}(t)$ . But we can still understand how the protein concentration changes over time by plotting the production-rate  $f(\hat{p})$  and the degradation-rate  $g(\hat{p})$  as functions of  $\hat{p}$ . From Eq. 44, we have

$$\frac{d\hat{p}}{d(t)} = \frac{\hat{V}\hat{p}^n}{1 + \hat{p}^n} - \hat{p} \quad (45a)$$

$$= f(\hat{p}) - g(\hat{p}) \quad (45b)$$

and thus

$$f(\hat{p}) = \frac{\hat{V}\hat{p}^n}{1 + \hat{p}^n} \quad g(\hat{p}) = \hat{p} \quad (46)$$

Note that when  $d\hat{p}/dt = 0$ , the concentration of the protein does not change over time (i.e.,  $\hat{p}$  remains constant over time). In other words, this would be a **steady-state gene expression**. This occurs if and only if the graphs of  $f(\hat{p})$  and  $g(\hat{p})$  intersect. There are different ways of these two graphs to intersect depending on the values of the two constants:  $\hat{V}$  and  $n$ . Let's take a look at these different ways:

**Case 1: Monostable (gene expression is stably off)  $V=1, n=1$ :** Plotting the  $f(\hat{p})$  and  $g(\hat{p})$  (see Fig. 6) we see that the production and degradation rate curves intersect only when  $\hat{p}=0$  (note that we relabeled  $\hat{p}$  as  $x$  in Fig. 6 for convenience). This means that the steady-state gene-expression is one in which no protein is produced from it. Note that when  $\hat{p} > 0$ , then the production rate is lower than the degradation rate (i.e., the  $f(\hat{p})$  curve lies below the  $g(\hat{p})$  curve in Fig. 6). Thus when the cell has some amount of the protein, the cell will degrade more than it produces the protein, driving the cell's protein concentration towards zero. Thus the gene-expression being zero (i.e.,  $\hat{p} = 0$ ) is a **stable equilibrium** - any perturbation away from this steady state value ( $\hat{p} = 0$ ) returns the protein concentration back to it. Given that there is only one steady-state and that it is stable, we call this a **monostable** case.

**Case 2: Monostable (gene expression is stably on)  $V=2, n=1$ :** Plotting the  $f(\hat{p})$  and  $g(\hat{p})$  (see Fig. 7) we see that the production and degradation rate curves intersect at two locations: At  $\hat{p}=0$  and at  $\hat{p} = 1$  (note that we relabeled  $\hat{p}$  as  $x$  in Fig. 7 for convenience). In Fig. 7, we see that  $f$  and  $g$  intersect at two locations. Thus there are two fixed points when  $V = 2$  and  $n = 1$ . One of the steady states corresponds to the gene being off (i.e.,  $\hat{p} = 0$ , white circle in Fig. 7). The other steady state corresponds to gene-expression at a non-zero value, specifically at  $\hat{p} = 1$  (black circle in Fig. 7). To determine the stability of these two steady states, we compare which of the two - either  $f$  or  $g$  - is greater than the other. When  $\hat{p}$  is between zero and one, we see that the production rate  $f$  is higher than the degradation rate  $g$  (i.e., the blue curve lies above the red curve in Fig. 7). Hence, when the protein concentration is a little above zero (i.e.,  $0 < \hat{p} < 1$ ), the more protein is produced than degraded, causing the cell increase the protein concentration even more (i.e.,  $d\hat{p}/dt > 0$  in Eq. 45b). Thus the protein concentration does not return to zero once the cell has even a few copies of the protein inside it. Hence we call this equilibrium state to be **unstable equilibrium**. When  $\hat{p} > 1$ , we see that the degradation rate is higher than the production rate (i.e., the blue curve lies below the red curve in Fig. 7) and thus the protein concentration would decrease towards  $\hat{p} = 1$  (i.e.,  $d\hat{p}/dt < 0$  in Eq. 45b). When  $\hat{p} < 1$ , we see that the degradation rate is lower than the production rate (i.e., the blue curve lies above the red curve in Fig. 7) and thus the protein concentration would increase towards  $\hat{p} = 1$  (i.e.,  $d\hat{p}/dt > 0$  in Eq. 45b). Thus any perturbation away from this equilibrium state,  $\hat{p} = 1$ , causes the protein concentration to return to  $\hat{p} = 1$ . For this reason, like in case 1 above, we say that this steady state is a stable equilibrium and that the gene expression can be stably on. Gene expression being "on" means that a non-zero value of the protein



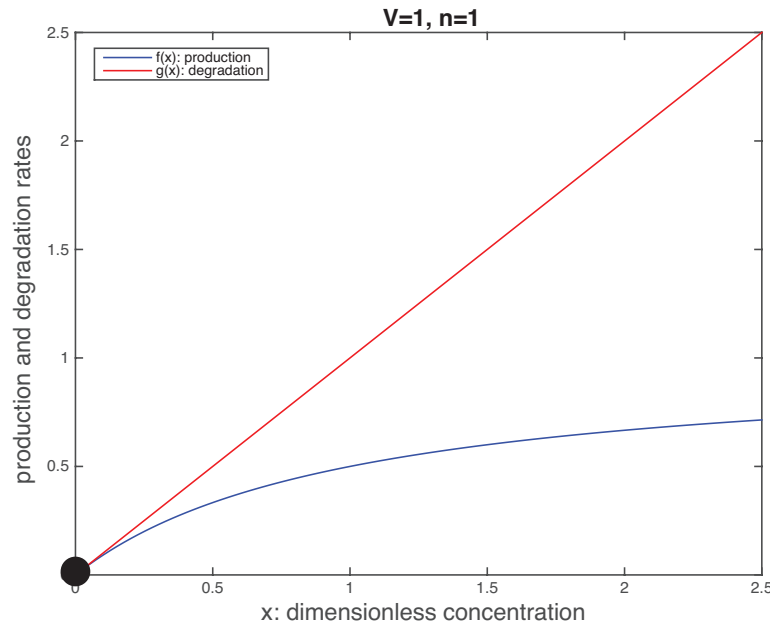


Figure 6: **Monostable (gene expression is stably off) for autoregulatory positive feedback:** Plot of the production-rate function  $f(x)$  (blue curve) and degradation-rate function  $g(x)$  (red curve) as defined by Eq. 46. For convenience, we have relabeled  $\hat{p}$  as  $x$ .  $\hat{V} = 1$  and  $n = 1$ . The black circle shows the only point where  $f$  and  $g$  intersect.

concentration can be stably maintained inside the cell. We still have a monostable case here because there is only one stable equilibrium state even though there are two equilibrium states.

**Case 3: Bistable (gene expression can be both stably off and stably on)  $V=2.5$ ,  $n=2$ :** Plotting the  $f(\hat{p})$  and  $g(\hat{p})$  (see Fig. 8) we see that the production and degradation rate curves now intersect at three locations: At  $\hat{p}=0$ ,  $\hat{p}=0.5$ , and  $\hat{p} = 2$  (note that we relabeled  $\hat{p}$  as  $x$  in Fig. 8 for convenience). Thus there are three steady states here. Using the argument given in case 2, we can deduce that  $\hat{p}=0$  and  $\hat{p} = 2$  are stable equilibria whereas  $\hat{p} = 0.5$  is an unstable fixed point. Thus, in this case, the gene expression can be stably off (i.e.,  $\hat{p} = 0$ ) or stably on (i.e.,  $\hat{p} = 2$ ). Using the style of argument given in case 2, we can further deduce that when the protein concentration  $\hat{p}$  is above 0.5 (the unstable equilibrium value), then the gene turns "on" (i.e.,  $\hat{p}$  evolves to and settles down at two) whereas if it is below 0.5, then the gene turns "off" (i.e.,  $\hat{p}$  evolves to and settles down at zero). For this reason, we call this a **genetic switch**: Like a light switch, the gene can either be off or on. The unstable equilibrium acts as the mid-position in a light switch. It is very difficult to put the light switch in the mid-position and any perturbation (little push) on the light switch at the mid-position will lead to either switching on or off the light. In retrospect, we also see that this is why we mentioned earlier that as the Hill coefficient increases, the more switch-like the gene-regulation becomes. We used  $n = 2$  here whereas in cases 1 and 2, the Hill coefficient was at its lowest possible value,  $n = 1$ .

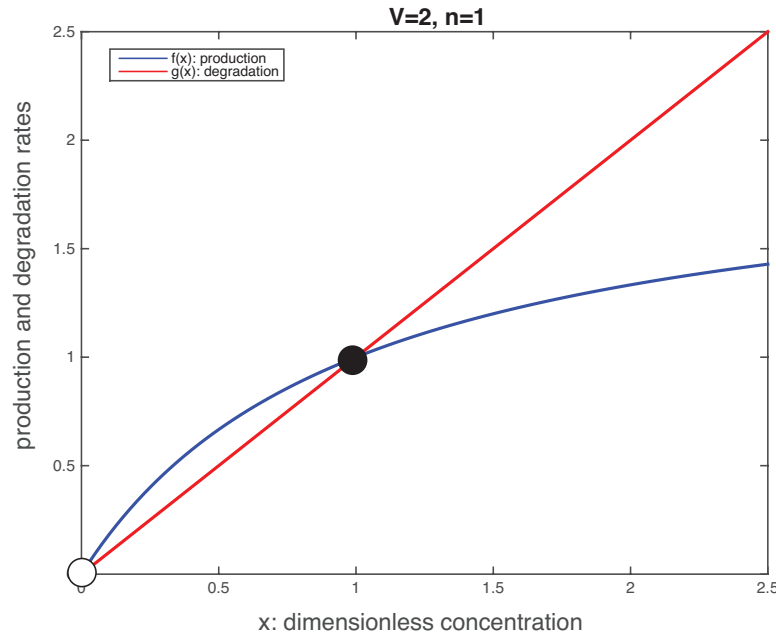


Figure 7: **Monostable (gene expression is stably on) for autoregulatory positive feedback:** Plot of the production-rate function  $f(x)$  (blue curve) and degradation-rate function  $g(x)$  (red curve) as defined by Eq. 46. For convenience, we have relabeled  $\hat{p}$  as  $x$ .  $\hat{V} = 2$  and  $n = 1$ . The black point, where  $f$  and  $g$  intersect, is a stable equilibrium point. The white point, where  $f$  and  $g$  intersect, is an unstable equilibrium point.

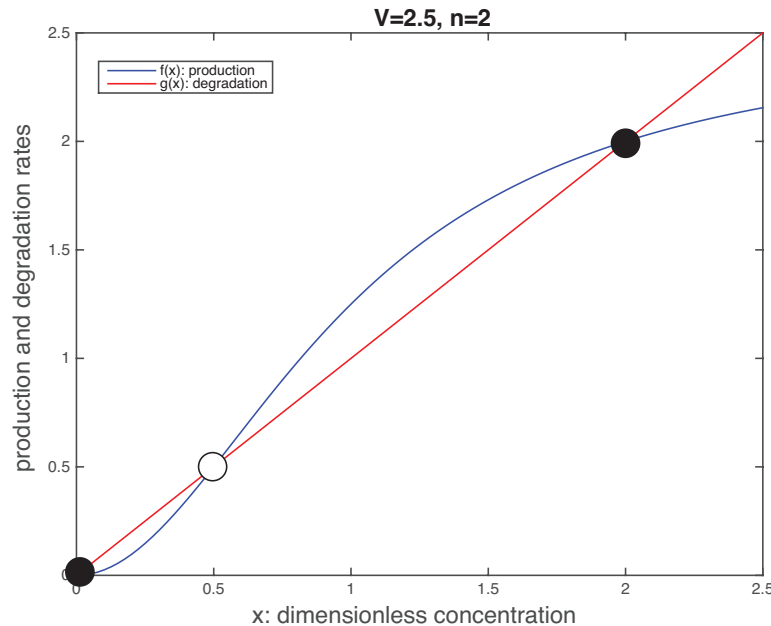


Figure 8: **Bistable gene-expression (gene expression can be stably off and stably on) for autoregulatory positive feedback:** Plot of the production-rate function  $f(x)$  (blue curve) and degradation-rate function  $g(x)$  (red curve) as defined by Eq. 46. For convenience, we have relabeled  $\hat{p}$  as  $x$ .  $\hat{V} = 2.5$  and  $n = 2$ . The two black points, where  $f$  and  $g$  intersect, are stable fixed points. The white point, where  $f$  and  $g$  intersect, is an unstable fixed point.

## 5 Gene-regulation scheme III: Negative feedback

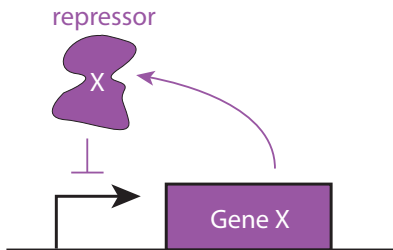


Figure 9: **Auto-regulatory negative feedback loop:** A gene X encodes a protein that is a repressor for the transcription of itself. Thus, the more X there is inside the cell, the smaller the production rate of X.

Just as in the case of positive feedback, we use a phenomenological equation to describe the opposite of positive feedback loop, namely the negative feedback loop. Here, the idea is that a gene X produces a protein X that **represses** its own production (Fig. 9). We call such a protein a **repressor**. Activator (in positive feedback) and repressor (in negative feedback) are both called **transcription factors** because they regulate the transcription of a gene (in their case, they regulate themselves). The idea behind the negative feedback is that the more protein X there is, the more of it will bind to the promoter of gene X. The more protein X is bound to the promoter of gene X, the lower the transcription rate, and thus translation rate of X. The phenomenological equation that we use to model negative feedback is

$$\frac{dp}{dt} = \frac{V}{1 + p^n} - p \quad (47)$$

where we have already taken the liberty of using the normalized units of time and concentration that we introduced for the positive feedback scenario (here we have dropped the "hats" on top of  $t$  and  $p$  for convenience). Note that now, the production rate of the protein is maximum ( $V$ ) when  $p = 0$ , which makes sense for the negative feedback scheme described above. As the  $p$  increases, we see that the production rate decreases. As  $p$  approaches infinity, we see that the production rate goes to zero. Note that Eq. 47 enables only one fixed point and that it is a stable fixed point (you can see this graphically by plotting the production rate and the degradation rate together as functions of  $p$ ).

A more interesting scenario occurs when we have two genes, X and Y, that repress each other (Fig. 10). Specifically, consider a gene-regulation scheme governed by the following equation:

$$\frac{dx}{dt} = \frac{V_1}{1 + y^n} - x \quad (48a)$$

$$\frac{dy}{dt} = \frac{V_2}{1 + x^n} - y \quad (48b)$$

where  $x$  and  $y$  are concentrations of protein X and Y respectively. Here, protein X is the repressor of gene Y and protein Y is the repressor of gene X. To find the steady state expression levels of genes X and Y, we can use the idea of **nullclines** from dynamical systems theory. Let  $U(x, y) = dx/dt$  (Eq. 48a) and  $V(x, y) = dy/dt$  (Eq. 48b). Then at steady state expression levels  $(x_0, y_0)$ , we must have  $U(x_0, y_0) = 0$  and  $V(x_0, y_0) = 0$ . In other words, if we consider all values  $(x, y)$  for which  $U(x, y) = 0$ , we would obtain a curve  $y = A(x)$ . By considering all points  $(x, y)$  for which  $V(x, y) = 0$ , we would obtain another curve

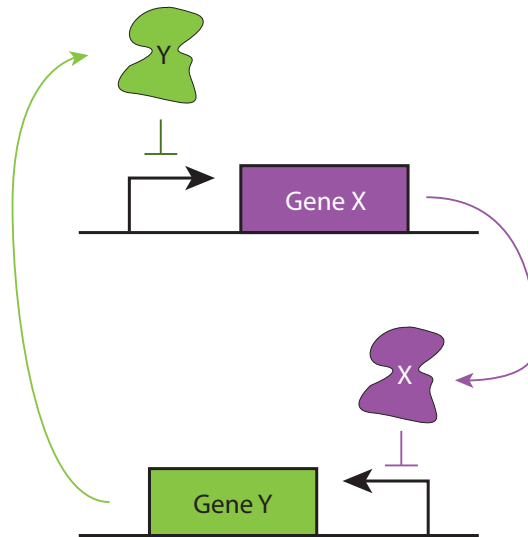


Figure 10: **Genetic circuit with two genes that repress each other:** X is a repressor of gene Y whereas Y is a repressor of gene X. This is sometimes called a **flip-flop switch**.

$y = B(x)$ . By plotting  $A$  and  $B$  as functions of  $x$ , if we find that the two curves intersect, then we would conclude that the location of the intersection,  $(x_0, y_0)$ , corresponds to the fixed point of the system. As an example, let us consider

$$\frac{dx}{dt} = \frac{6}{1+y} - x \quad (49a)$$

$$\frac{dy}{dt} = \frac{3}{1+x} - y \quad (49b)$$

Then setting  $U = 0$  and  $V = 0$ , we find the following nullclines:

$$y = \frac{6}{x} - 1 \quad (\text{satisfies } U(x, y) = 0) \quad (50a)$$

$$x = \frac{3}{y} - 1 \quad (\text{satisfies } V(x, y) = 0) \quad (50b)$$

Sketching by hand Eq. 50a and Eq. 50b together in one place, we graphically see that the two nullclines intersect at only one location,  $(x_0, y_0)$  (Fig. 11). In fact, we do not even need to know the specific values of  $x_0$  and  $y_0$ . Furthermore, we note that the intersection of the two nullclines divide the  $xy$ -plane into four regions. By looking at Eq. 49a and Eq. 49b change at each of the four regions, we deduce that the fixed point  $(x_0, y_0)$  is in fact a stable fixed point.

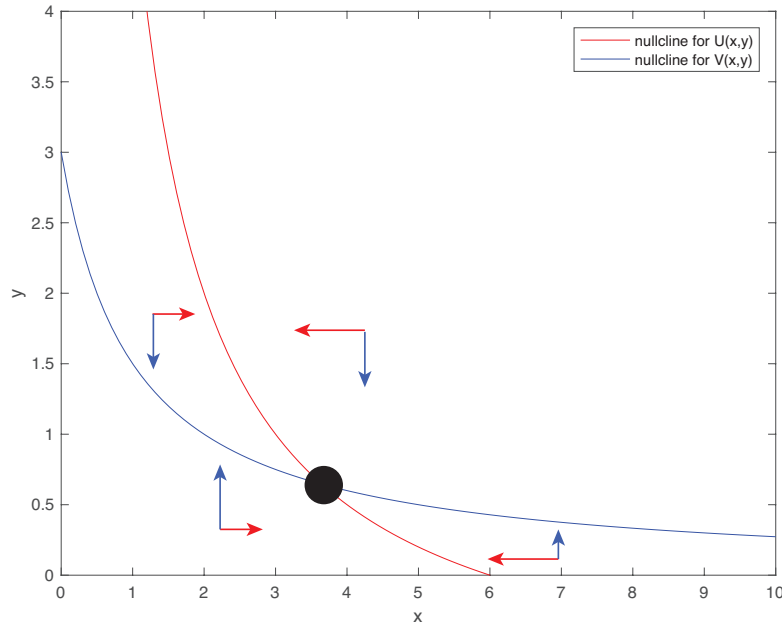


Figure 11: **Nullclines dictated by Eq. 50a and Eq. 50b** Black dot represents a fixed point. By analyzing whether Eq. 49a and Eq. 49b are positive or negative in each of the four regions that are created by the two nullclines, we deduce that the fixed point is stable.

## 6 Why is gene expression stochastic?

Earlier in this chapter, we discussed equations that describe basic gene-regulatory schemes. The equations that we discussed above were deterministic equations - They were ordinary differential equations whose solutions are deterministic. This means that if we know the concentration of the relevant molecules (mRNA or proteins) in a cell at a particular time point, then the equation tells us what the concentrations of those molecules were in the past and what they will be in the future. Information about the present state is all that one would need to exactly specify the past and the future of the cell. This "clockwork" or "mechanical" view of gene expression and cellular processes in general (such as motion of bacteria), is in fact, not entirely accurate. As we will see below, cellular processes such as gene expression in a cell are stochastic. The typical reason that is often given in the literature for this is that there are very few copies of a molecule involved in typical cellular processes, such as the number of mRNA or protein produced from a gene inside a cell. As an example, a cell may contain between one to a hundred copies of an mRNA from a given gene. Dividing 1 to 100 mRNA molecules by the volume of a cell, say  $V$ , gives us:  $1/V, 2/V, \dots, 100/V$ . We see that the concentration is not continuous, but rather is discretized in steps of  $1/V$ . Ordinary differential equations from earlier in this chapter fail to capture these discrete changes in concentration (as they also failed to capture stochastic cell division and death - see the Appendix to Chapter 3). But intuition tells us that if  $1/V$  is "small enough", then we can "smooth out" the discreteness. As we will see, the "small enough" quantitatively means that  $1/V$  is much smaller than the average concentration. In other words, when there are large number of copies of a molecule (mRNA or protein from a gene) inside a cell, we can use the deterministic equations that we derived earlier in this chapter. This intuitively makes sense because if we have 1000,000 molecules, then losing or gaining 1 or 2 molecules doesn't make much difference.

But as we discuss in the Appendix to Chapter 3, there is a deeper reason for why gene expression is

stochastic. If we watch a single cell over time, we cannot predict exactly when it will transcribe a mRNA from a given gene and when it will translate a protein from each mRNA. This is because we lack the necessary information such as exactly when each RNA polymerase will bind to the DNA, the location and movement of each copy of mRNA from the gene of interest, the location and movement of each ribosome, and many others. The key here is that the amount of information that we would need to construct the movement of all the key molecules inside the cell by using Newton's laws of motion and any other laws of physics and chemistry would be huge. Thus, even if we could see the location of every ribosome and RNA polymerase inside a cell, we would need to know the location of all other molecules inside the cell that collide with them in order to do better than describing their motion as diffusion (the later being a probabilistic description). For the same reason that we can, at best, only give a stochastic description of cell division and death (see the Appendix to Chapter 3), we can only give, at best, a stochastic description of gene expression - the number of copies of mRNA and protein from a given gene over time is a stochastic quantity. As we will see, the stochastic description approaches a deterministic description as the number of copies of the molecule approaches a sufficiently "large" number (to be quantified below).

Below, we will start by building two standard approaches for modelling **stochastic gene expression**. We will then describe the stochastic dynamics of the standard forms of gene circuits - constitutive circuit, positive feedback, negative feedback.

## 7 Master equation

In the Appendix to Chapter 3, we derive the **Master equation** by using stochastic birth and death of cells. But if we replace "cells" by "molecules", "birth of a cell" by a "creation of a molecule in a cell", and "death of a cell" by "degradation of a molecule in a cell", then we obtain the same Master equation to describe the number of molecules inside a cell. Specifically, let  $n$  be the number of copies of a molecule (e.g., mRNA or protein from gene X). Then the probability  $P_n(t)$  that there are  $n$  copies of the molecule in a cell at time  $t$  is given by the following Master equation

$$\frac{dP_n}{dt} = f_{n-1}P_{n-1} + g_{n+1}P_{n+1} - g_nP_n - f_nP_n \quad (51)$$

where  $f_n dt$  is the probability that a cell with  $n$  copies of the molecule will create another copy of the molecule in time interval  $(t, t + dt)$ , and  $g_n dt$  is the probability that a cell with  $n$  copies of the molecule will degrade a copy of the molecule in time interval  $(t, t + dt)$ . We can pictorially represent the stochastic creation and degradation of the molecules as a Markov chain (Fig. 12).

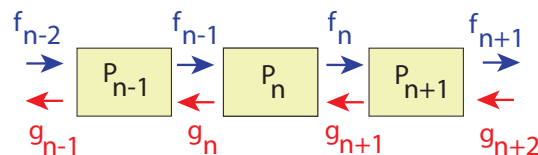


Figure 12: A **Markov chain**: Visualisation of stochastic creation and degradation of molecules in a cell.  $n$  is the number of copies of a molecule (e.g., mRNA, protein) inside the cell.  $f_n dt$  to be the probability that a cell with  $n$  copies of the molecule creates another copy of the molecule within an infinitesimal time interval  $dt$ .  $g_n dt$  is the probability that a cell with  $n$  copies of the molecule degrades one copy of the molecule within an infinitesimal time interval  $dt$ .

To experimentally measure the probability  $P_n$  for each  $n$ , we can try two methods. Suppose that there are 1000 genetically identical cells. In one method, we would use a camera that is attached to a microscope to take a snapshot of each of the 1000 cells at the same time, and then count the number of copies of

the mRNA in each cell that is transcribed from a gene of interest. We can then plot the histogram of the number of copies of the mRNA per cell. This histogram estimates the  $P_n(t)$  for every  $n$  at a given time  $t$ . This estimate would approach a "true" underlying probability distribution  $P_n(t)$  as the number of cells increases from 1000 to infinity. But we do not need to know the exact probability distribution - a simple histogram would still tell us the amount of cell-to-cell variability in the number of copies of the mRNA. A second method would involve focusing on a single cell with a microscope and then recording the number of copies of the mRNA and/or the protein in that cell by following it over time. This would result in a time-lapse movie in which we record the birth and death of every mRNA and/or protein from a single gene. We can then count the copy number of mRNA and/or protein from a gene in that cell at each timeframe of the movie. After filming the cell for a sufficiently long enough time (e.g., until after it divided many times), we can pool together the measurements from all the timeframes to make one histogram of the copy number of mRNA and/or protein. Such a histogram would not tell us  $P_n(t)$  but rather the time-independent, **steady-state probability distribution**  $P_n$ . In fact, the steady-state probability distribution is typically more important than knowing the  $P_n(t)$  at every time  $t$ . To see this, note that the cells were already growing (and living) before we switch on the microscope. This means that the cells likely already reached a steady-state distribution before you even started the experiment. An exception occurs for genes whose expression does not begin until we tell them to - this occurs, for example, for inducible genes for which we can add an inducer (i.e., chemical that activates gene expression) to the growth medium of the cells right after switching on the microscope.

We next turn to analyzing the Master equation (Eq. 51) for the standard gene regulations - (1) constitutive, (2) auto-regulatory positive feedback, and (3) negative feedback. Throughout, we will let  $n$  be the number of copies of a protein. For a constitutive gene expression, we have  $f_n = k$ . For an auto-regulatory positive feedback, we can let  $f_n = Vn^h/(K^h + n^h)$  where  $h$  is the Hill coefficient. For an auto-regulatory negative feedback, we can let  $f_n = VK^h/(K^h + n^h)$ . In all three cases, we will consider a simple degradation scheme, in which a molecule degrades independently of the state of the other molecules and with a constant probability of degradation per unit time (i.e.,  $g_n = \gamma n$ , where  $\gamma$  is a positive constant).

The positive-feedback rate above assumes that the gene produces nothing when  $n = 0$ . We will see that this matters for its steady-state distribution. To let the gene make a protein even when none is present, we can add a constant **basal production rate**  $\epsilon > 0$ :  $f_n = \epsilon + Vn^h/(K^h + n^h)$ . We will compare these two models in Section 8.2.

## 7.1 Constitutive gene expression - Master equation

We are interested in two quantities: (1) the average copy number  $\langle n \rangle$  of the mRNA or protein in a cell, and (2) the steady-state distribution  $P_n$  (i.e., what  $P_n(t)$  becomes after a long time). We will now compute both for the case of a constitutive gene expression.

### 7.1.1 Average number of copies of a molecule in a cell

This calculation proceeds the same way as in computing the average number of cells with the Master equation in the Appendix to Chapter 3. By the definition of averages, we have

$$\frac{d\langle n \rangle}{dt} = k(\langle n \rangle + 1) + \gamma \left[ \sum_{n=0}^{\infty} n(n+1)P_{n+1}(t) \right] - k \sum_{n=0}^{\infty} nP_n(t) - \gamma \sum_{n=0}^{\infty} n^2P_n(t) \quad (52)$$

We can simplify Eq. 52 by using a mathematical trick:  $n = (n+1) - 1$  and  $n = (n-1) + 1$ . We can also simply Eq. 52 by using the fact that  $P_{-1}(t) = 0$  and that  $\sum P_n(t) = 1$ . Then Eq. 52 becomes

$$\frac{d\langle n \rangle}{dt} = k - \gamma \langle n \rangle \quad (53)$$

Note that Eq. 53 has the same form as the equation for the constitutive gene expression scheme that we derived earlier in this chapter (Eq. 9). The only difference is in the meaning. Eq. 9 was for the protein concentration " $p$ ", which we inherently assumed to be deterministic. In other words, we assumed that every cell with the same protein concentration now will have the same concentration at every future instance. On the other hand, Eq. 53 describes the average copy number  $\langle n \rangle$  across a population of cells rather than the concentration in any single cell. In this probabilistic view, two cells with the exactly the same copy number  $n$  of a mRNA can have vastly different copy number of mRNA in the future. But the average copy number  $\langle n \rangle$  of the mRNA from the same gene would deterministically change over time (Eq. 53). In particular, we have

$$\langle n(t) \rangle = \frac{k}{\gamma}(1 + n_0 e^{-\gamma t}) \quad (54)$$

where  $n_0$  is a constant that depends on the initial condition (e.g.,  $n_0 = -1$  if  $\langle n(t=0) \rangle = 0$ ).

### 7.1.2 Steady-state solution of the Master equation

In steady-state, we have  $dP_n/dt=0$  for all values of  $n$ . Thus the Master equation (Eq. 51) becomes

$$0 = f_{n-1}P_{n-1} + g_{n+1}P_{n+1} - g_nP_n - f_nP_n \quad (55)$$

In particular, for the constitutive gene expression scheme, we have

$$0 = kP_{n-1} + \gamma(n+1)P_{n+1} - \gamma nP_n - kP_n \quad (56)$$

Rearranging the terms in Eq. 56, we obtain

$$-\gamma(n+1)P_{n+1} + kP_n = -\gamma nP_n + kP_{n-1} \quad (57)$$

Above equation is true for all values of  $n$  ( $n \geq 0$ ). Eq. 57 generates a chain of equalities, starting from  $n = 0$  and moving up to larger values of  $n$ . At the lowest end of this chain (i.e., at  $n = 0$ ), Eq. 57 and the fact that  $P_{-1}=0$  yields

$$-\gamma P_1 + kP_0 = 0 \quad (58)$$

Thus, the chain of equalities (Eq. 57) tells us that

$$-\gamma nP_n + kP_{n-1} = 0 \quad (59)$$

for all values of  $n$ . Now, with Eqs. 58 and 59, we can solve for  $P_n$ . To see this, we first note that Eq. 58 gives us

$$P_1 = \frac{k}{\gamma}P_0 \quad (60)$$

Then, recursively applying Eq. 59 from  $n > 0$  down to  $n = 0$ , we obtain



$$P_n = \left(\frac{k}{\gamma}\right)^n \frac{P_0}{n!} \quad (61)$$

which reproduces Eq. 60 when  $n = 1$ . Eq. 61 tells us that if we know what  $P_0$  is, then we know what  $P_n$  is for any value of  $n$ . To determine  $P_0$ , we use the fact that summing all the probabilities yields 1. We thus have

$$1 = \sum_{n=0}^{\infty} P_n \quad (62a)$$

$$= P_0 \sum_{n=0}^{\infty} \left(k/\gamma\right)^n \frac{1}{n!} \quad (62b)$$

$$= P_0 e^{k/\gamma} \quad (62c)$$

In Eq. 62c, we used the fact that the summation in Eq. 62b is, by definition, the Taylor expansion of an exponential. From Eq. 62c, we have  $1 = P_0 e^{k/\gamma}$ , or in other words,

$$P_0 = e^{-k/\gamma} \quad (63)$$

Note that  $P_0$  is a constant (i.e., does not depend on time) because we have computed steady-state probabilities. Plugging Eq. 63 into Eq. 61, we obtain

$$P_n = \left(\frac{k}{\gamma}\right)^n \frac{e^{-k/\gamma}}{n!} \quad (64)$$

This is the steady state distribution of  $n$  for a constitutive gene expression (Fig. 13). It is a ubiquitous form of probability distribution called the **Poisson distribution**. The Poisson distribution has the following general form

$$P(n) = (\langle n \rangle)^n \frac{e^{-\langle n \rangle}}{n!} \quad (65)$$

An important property of the Poisson distribution (Eq. 65) is that a random variable  $n$  follows the Poisson distribution, then its mean  $\langle n \rangle$  and variance  $\langle (n - \langle n \rangle)^2 \rangle$  are the same. From the steady-state distribution of  $n$  (Eq. 64), we see that the mean copy number of the molecule, say the mRNA from a gene, is  $k/\gamma$ . Thus, we have

$$\langle n \rangle = \langle (n - \langle n \rangle)^2 \rangle = \frac{k}{\gamma} \quad (66)$$

Note that Eq. 66 is precisely the steady-state copy number that we would obtain from the deterministic equation for  $\langle n \rangle$  (Eq. 54). Note that the steady-state copy number  $\langle n \rangle$  in Eq. 54 corresponds to the value that  $\langle n \rangle$  approaches after a long time (i.e.,  $t \rightarrow \infty$ ). Furthermore, the **fractional error**, which is the standard deviation in  $n$  (i.e.,  $\sqrt{\langle (n - \langle n \rangle)^2 \rangle}$ ) divided by the average copy number  $\langle n \rangle$  is

$$\frac{\sqrt{\langle (n - \langle n \rangle)^2 \rangle}}{\langle n \rangle} = \frac{1}{\sqrt{\langle n \rangle}} \quad (67)$$

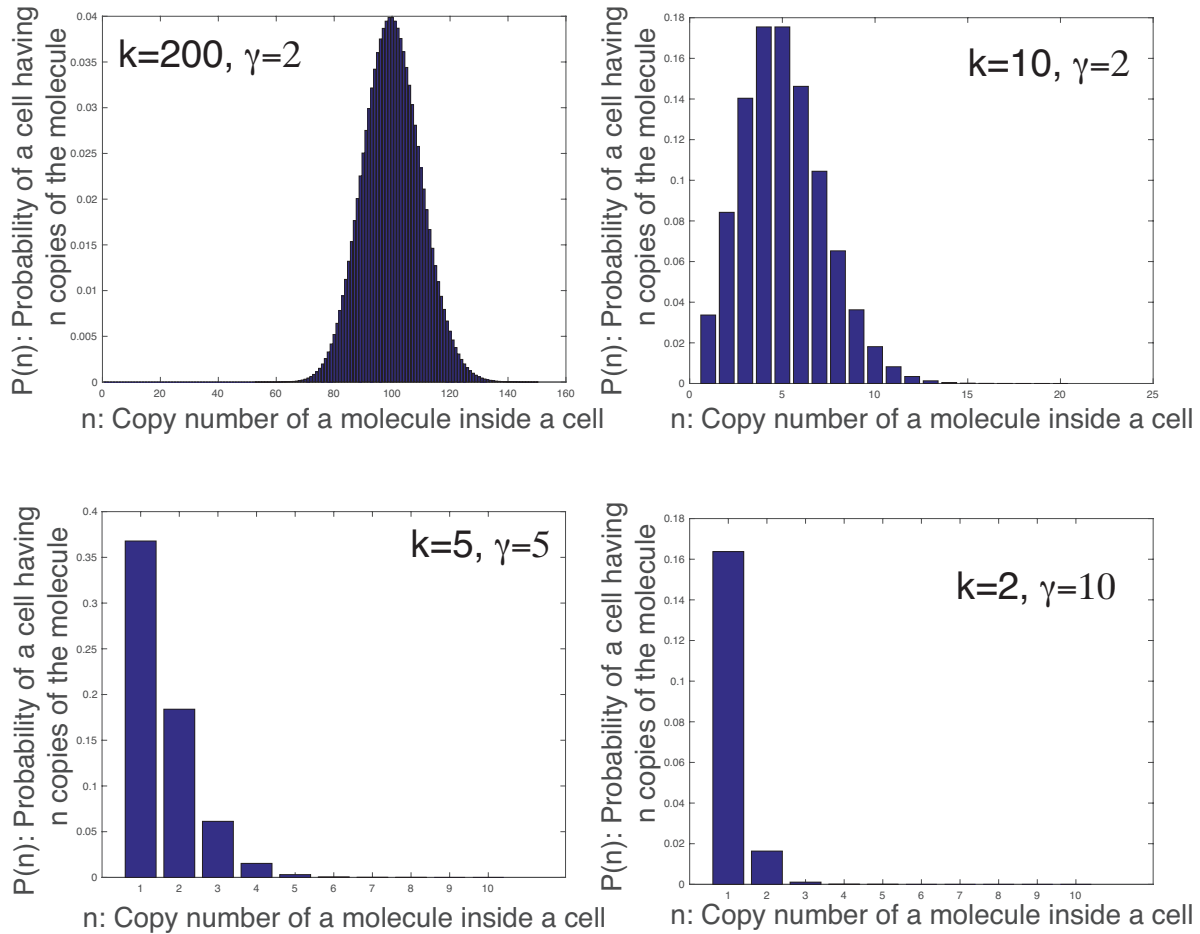


Figure 13: **Steady-state probability distributions for a constitutive gene expression.** Histograms of  $P(n)$  given by Eq. 64 for different values of the production rate  $k$  and degradation constant  $\gamma$ .

Thus, as the copy number of the mRNA (or protein) in a cell increases, the fractional error decreases. The fractional error (Eq. 67) quantifies how important the stochastic dynamics is for the gene of interest. If the cell has a low number of copies of the mRNA (or protein), then the stochastic effect is non-negligible because the average deviation that a cell has from the population-level average is large. This is why one often says that stochastic gene expression is due to a low copy number of the mRNA and/or the protein. But as we discussed at the beginning of this lecture, the deeper answer is that we never have all the necessary information to exactly predict the cell's gene-expression dynamics at every point in time. Although we derived Eq. 67 only for a constitutive gene expression, we would obtain similar outcomes for other types of gene-regulatory schemes.

### 7.1.3 Simulating the Master equation

We can simulate the stochastic gene-expression dynamics by using a type of Monte Carlo-simulation algorithm called the **Gillespie algorithm** (Fig. 14). The basic idea behind the Gillespie algorithm is that we can simulate the transitions between the different "boxes" in the Markov chain (Fig. 12) by using the  $f_n$  to calculate the time that we need to wait for the creation of another copy of the mRNA (or protein) and using the  $g_n$  to calculate the time that we need to wait for the degradation of one copy of the mRNA or the protein. The Gillespie algorithm works for almost any form of gene-regulatory circuits.

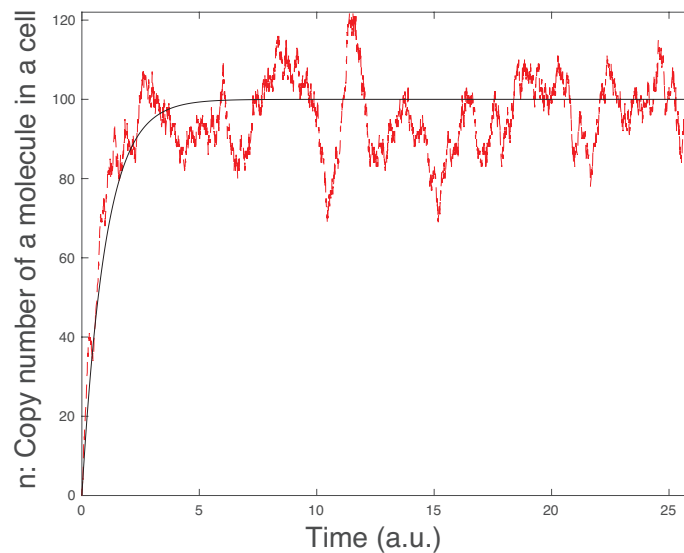


Figure 14: **Using the Gillespie algorithm to simulate the stochastic, constitutive gene expression that is dictated by the Master equation (Eq. 51).**  $k = 100$  and  $\gamma = 1$ , where  $f_n = k$  and  $g_n = \gamma n$  in Eq. 51. Red curve shows the stochastically changing copy number  $n$  of the mRNA in a cell as a function of time (computed by the Gillespie algorithm). Black curve is the deterministically changing mean  $\langle n \rangle$  (computed using Eq. 53).

## 8 Fokker-Planck equation

Our second approach for modeling stochastic gene expression uses the **Fokker-Planck equation**. The main idea behind the Fokker-Planck equation is that we approximate the Master equation by turning its discrete values of  $n$  into continuous values of  $n$ . To see how this modification can help us, note that the Master equation is difficult to solve, even for just the set of steady-state probabilities  $\{P_n\}$ , because there

are infinitely many  $P_n$ 's, with each  $P_n$  depending on the other  $P_n$ 's. As we will see, the Fokker-Planck equation results from turning the infinite number of equations for the infinite set  $\{P_n\}$  into a single equation - an equation for  $P(n)$ , where  $n$  is now a continuous variable.

To turn the discrete values of  $n$  into continuous values, we will use the following technique. Let  $\Delta n$  be some "typical" variation in the copy number  $n$ . For concreteness, we can let  $\Delta n$  be the standard deviation in  $n$ . Now suppose that  $\Delta n/\langle n \rangle$  is sufficiently small. For a constitutive gene expression, by taking  $\Delta n$  to be the standard deviation in  $n$ ,  $\Delta n/\langle n \rangle$  is sufficiently small means that the fractional error,  $1/\sqrt{\langle n \rangle}$ , is sufficiently small according to Eq. 67. Consider some arbitrary, smooth function  $L(n)$ . Then the Taylor expansion of  $L(n)$  around any value  $n_0$  is

$$L(n_0 + \Delta n) = L(n_0) + \frac{\partial L}{\partial n} \Delta n + \frac{1}{2} \frac{\partial^2 L}{\partial n^2} (\Delta n)^2 + \dots \quad (68)$$

where the derivatives are evaluated at  $n_0$  and  $\Delta n$  is a typical deviation in  $n$ . Note that here we are assuming that  $n$  is a continuous variable. If  $\Delta n$  is sufficiently small, we can ignore all the terms in Eq. 68 that are beyond the second order term. Now, the main idea that we use to derive the Fokker-Planck equation is that if  $\Delta n/\langle n \rangle$  is small, then we can just boldly Taylor expand all the functions of  $n$  in the Master equation (Eq. 51) up to and including second order in  $n$  as done in Eq. 68. Specifically, using the Taylor expansion, we obtain

$$f(n-1)P(n-1) \approx f(n)P(n) - \frac{\partial}{\partial n}(f(n)P(n)) + \frac{1}{2} \frac{\partial^2}{\partial n^2}(f(n)P(n)) \quad (69a)$$

$$g(n+1)P(n+1) \approx g(n)P(n) + \frac{\partial}{\partial n}(g(n)P(n)) + \frac{1}{2} \frac{\partial^2}{\partial n^2}(g(n)P(n)) \quad (69b)$$

In these two expansions, the shift is one molecule,  $\Delta n = 1$ . This is the size of a single reaction step, not the standard deviation introduced above. The approximation requires  $f(n)P(n)$  and  $g(n)P(n)$  to vary smoothly over that one-molecule step. A small fractional error by itself does not guarantee this condition, especially near  $n = 0$ . Substituting Eqs. 69a and 69b into the Master equation (Eq. 51) and then ignoring all terms that involve third and higher-order derivatives, we obtain

$$\frac{\partial P(n,t)}{\partial t} = -\frac{\partial}{\partial n} \left\{ (f(n) - g(n))P(n) - \frac{1}{2} \frac{\partial}{\partial n} [(f(n) + g(n))P(n)] \right\} \quad (70)$$

This is the **Fokker-Planck equation**. It is a continuous version of the Master equation in which we treat the  $n$  as a continuous variable. The term inside the curly brackets in Eq. 70 is called the **net probability flux**, which we denote by  $J(n,t)$ . We can thus rewrite Eq. 70 more compactly as

$$\frac{\partial P(n,t)}{\partial t} = -\frac{\partial J(n,t)}{\partial n} \quad (71)$$

Note that Eq. 71 is just the **transport equation** in one dimension. To understand the meaning of the net probability flux  $J(n,t)$ , consider the Markov chain picture (Fig. 12) in which we consider a chain of boxes, each representing the copy number of the mRNA or protein that a cell can have. We said earlier that we can view  $P_n$  as the fraction of genetically identical cells that have  $n$  copies of the mRNA or the protein. To visualize this pictorially, we can envision "putting" cells into a box labeled  $n$  if those cells have  $n$  copies of the molecule at a given time  $t$ . Then at a later time  $t + dt$ , we will have to transfer some cells to the box labeled  $n - 1$  (to the left of the box labeled  $n$  in Fig. 12) if those cells have degraded a molecule during time interval  $(t, t + dt)$  and transfer some cells to the box labeled  $n + 1$  (to the right of

the box labeled  $n$  in Fig. 12) if those cells have created a molecule during time interval  $(t, t + dt)$ . The transfer to the box labeled  $n - 1$  represents a probability flux to the left (let's denote this as  $J_-$ ) while the transfer to the box labeled  $n + 1$  represents a probability flux to the right (let's denote this as  $J_+$ ). Then the net probability flux  $J(n, t)$  is  $J_+ - J_-$ . According to the Fokker-Planck equation,  $n$  is now continuous. This means that we have a continuous line of  $n$ -values, at that there is a net current  $J(n, t)$  at each point on this line (each point representing a particular value of  $n$ ). In light of this interpretation, the Fokker-Planck equation (Eq. 71) is just a statement that if we sum up the probabilities  $P(n)$  over all values of  $n$ , then we should always get one.

## 8.1 Steady-state solution of the Fokker-Planck equation

Solving the Fokker-Planck equation (Eq. 71) by hand is difficult. But as was the case with the Master equation, finding the time-independent, steady-state probability  $P(n)$  of the Fokker-Planck equation is easier and one that we can do by hand. For the one-variable gene circuits considered here, we can write the zero-current steady-state probability density  $P(n)$  in terms of an integral. The Master equation also has a steady-state recurrence, but the continuous expression gives us another way to understand the distribution. In steady state, the Fokker-Planck equation (Eq. 71) becomes

$$\frac{\partial P(n, t)}{\partial t} = 0 \quad (72)$$

This means that the net probability current  $J(n, t)$  is a constant. In fact, it must be zero in steady state and for all values of  $n$ . The reason is that when  $n = 0$ , we must have  $J(0, t) = 0$  since no probability current can flow past  $n = 0$  either from  $n > 0$  into  $n < 0$  or from  $n < 0$  to  $n > 0$ . There are no cells with negative copy numbers of the mRNA or protein. Since  $J(n, t)$  is a constant (the same value at all  $n$ ) at steady state, we must indeed have  $J(n, t) = 0$  in steady-state. Thus, from the definition of  $J(n, t)$  (the curly bracket in Eq. 70), we have

$$(f(n) - g(n))P(n) = \frac{1}{2} \frac{\partial}{\partial n} [(f(n) + g(n))P(n)] \quad (73)$$

To simplify above equation, we define  $H(n) = (f(n) + g(n))P(n)$ . Then Eq. 73 becomes

$$\frac{f(n) - g(n)}{f(n) + g(n)} H(n) = \frac{1}{2} \frac{dH}{dn} \quad (74)$$

We can solve for  $H(n)$  in Eq. 74 by separating variables as follows:

$$2 \frac{f(n) - g(n)}{f(n) + g(n)} = \frac{1}{H(n)} \frac{dH}{dn} \quad (75)$$

Integrating Eq. 75 over  $n$  and realizing that the right side of Eq. 75 is just  $d \log(H)/dn$ , we obtain

$$H(n) = A \exp \left( \int_0^n 2 \frac{f(x) - g(x)}{f(x) + g(x)} dx \right) \quad (76)$$

where  $A$  is a constant to be determined. From the definition of  $H(n)$ , we can now solve for  $P(n)$ :

$$P(n) = \frac{A}{f(n) + g(n)} \exp \left( \int_0^n 2 \frac{f(x) - g(x)}{f(x) + g(x)} dx \right) \quad (77)$$

Eq. 77 is the **steady-state solution of the Fokker-Planck equation**. It holds for these one-variable models when a normalizable, zero-current steady-state density exists and  $f(n) + g(n) > 0$  on the interval where we use the formula. We did not assume specific functional forms for  $f$  and  $g$ . An absorbing boundary needs separate treatment. From Eq. 77, we see that  $A$  is a normalization factor that ensures that summing the probabilities yields one (i.e.,  $\int P(n)dn = 1$ ). The integral inside the exponential looks like a **potential energy function**. Motivated by this, let us define

$$U(n) = - \int_0^n 2 \frac{f(x) - g(x)}{f(x) + g(x)} dx \quad (78)$$

Then we can rewrite the steady-state solution of the Fokker-Planck equation as

$$P(n) = \frac{A}{f(n) + g(n)} \exp(-U(n)) \quad (79)$$

The main idea here is that  $\exp(-U(n))$  is like the Boltzmann factor from statistical mechanics.  $U(n)$  is like the energy measured in units of  $kT$ , where  $k$  is the Boltzmann constant and  $T$  is the temperature. In this case, there is no real "temperature" or the Boltzmann factor. So this is just an analogy. Nonetheless, this analogy gives us an intuition for what  $P(n)$  should look like for certain genetic circuits.

We must keep the factor  $1/(f(n) + g(n))$  as well as the exponential. Both change with  $n$ , so both affect where  $P(n)$  has its peaks. For example, taking the logarithm and then differentiating gives

$$\begin{aligned} \ln P(n) &= \ln \left( \frac{A}{f(n) + g(n)} \right) - U(n), \\ \frac{d \ln P}{dn} &= - \frac{f'(n) + g'(n)}{f(n) + g(n)} + 2 \frac{f(n) - g(n)}{f(n) + g(n)}. \end{aligned} \quad (80)$$

Here a prime means a derivative with respect to  $n$ . At an interior peak,  $dP/dn = 0$  and thus  $d \ln P/dn = 0$ . So a peak must satisfy

$$2(f(n) - g(n)) = f'(n) + g'(n). \quad (81)$$

We still need to check that the slope changes from positive to negative to identify a peak rather than a minimum. Note that the deterministic fixed points instead satisfy  $f(n) = g(n)$ . The two conditions are not the same. Thus, two stable deterministic fixed points, or two wells in  $U(n)$ , do not by themselves guarantee two peaks in  $P(n)$ .

## 8.2 Positive feedback needs a way to restart from zero

There is another issue that we must check before using the potential analogy for positive feedback. Consider the rates we wrote earlier:

$$f_n = \frac{Vn^h}{K^h + n^h}, \quad g_n = \gamma n. \quad (82)$$

If  $n = 0$ , then  $f_0 = g_0 = 0$ . The cell has no protein to activate the gene, so the gene cannot produce another protein. The cell also has nothing to degrade. Once the system reaches zero, it stays there. We call zero an **absorbing state**.

We can see what this does to the steady-state distribution directly from the Master equation. At  $n = 0$ , there is no inflow from  $n = -1$  and no degradation from zero. Therefore

$$\frac{dP_0}{dt} = g_1 P_1 - f_0 P_0 = \gamma P_1. \quad (83)$$

At steady state,  $dP_0/dt = 0$ , so  $P_1 = 0$ . Next, the steady-state equation at  $n = 1$  gives

$$0 = f_0 P_0 + g_2 P_2 - (f_1 + g_1) P_1 = 2\gamma P_2. \quad (84)$$

Thus  $P_2 = 0$ . Repeating the same argument gives  $P_3 = 0$ ,  $P_4 = 0$ , and so on. The probabilities must sum to one, so the only normalized steady-state distribution is

$$P_0 = 1, \quad P_n = 0 \quad \text{for every } n \geq 1. \quad (85)$$

This result is exact for the discrete model. We did not use the Fokker-Planck approximation.

What happened to the stable high-expression state that we found in the deterministic model? The deterministic equation describes a continuous concentration. Starting near the stable high state, it returns there after a small perturbation. But in the stochastic model, individual production and degradation events can take the system away from that state. It can eventually lose every protein. Production is bounded by  $V$ , while degradation grows as  $\gamma n$ , so this model cannot avoid absorption by continually growing to larger copy numbers. Starting with a finite copy number, it eventually reaches zero with probability one.

The wait can be extremely long. During a finite experiment, we may see cells remain near the high-expression state, and we may see a histogram with a low and a high peak. This does not contradict Eq. 85. Such a histogram need not be the infinite-time steady-state distribution. A long-lived state that the system eventually leaves is called a **metastable state**.

To let the gene restart after losing its last protein, we add basal production:

$$f_n = \epsilon + \frac{V n^h}{K^h + n^h}, \quad g_n = \gamma n, \quad \epsilon > 0. \quad (86)$$

The basal rate  $\epsilon$  describes production that does not require this protein to activate its own gene. In this one-variable model, we combine the underlying transcription and translation into a protein-production event. Both  $\epsilon$  and  $V$  have units of molecules per unit time. The maximum production rate is now  $\epsilon + V$ , not  $V$  alone.

Now  $f_0 = \epsilon$ . Even with no protein present, the cell can make a protein and leave zero. It waits an average time  $1/\epsilon$  for this first production event. A smaller basal rate makes this wait longer. Basal production does not force the gene into the high-expression state. The newly made protein might simply degrade. But the cell can try again. This is the missing process in Eq. 82.

We can also see the change mathematically. With basal production, the boundary equation at steady state is

$$0 = \gamma P_1 - \epsilon P_0, \quad P_1 = \frac{\epsilon}{\gamma} P_0. \quad (87)$$

Unlike before, steady state no longer forces  $P_1$  to vanish. Putting  $f_0 P_0 = g_1 P_1$  into the steady-state equation for  $P_1$  gives  $f_1 P_1 = g_2 P_2$ . Repeating this step gives

$$f_n P_n = g_{n+1} P_{n+1}, \quad \frac{P_{n+1}}{P_n} = \frac{\epsilon + V n^h / (K^h + n^h)}{\gamma(n+1)}. \quad (88)$$

This equality says that, at steady state, the probability flowing from  $n$  to  $n+1$  balances the probability flowing back. Iterating it gives

$$P_n = P_0 \prod_{j=0}^{n-1} \frac{\epsilon + V j^h / (K^h + j^h)}{\gamma(j+1)}, \quad n \geq 1. \quad (89)$$

We choose  $P_0$  so that  $\sum_{n=0}^{\infty} P_n = 1$ . This is possible because production remains bounded while degradation grows with  $n$ . More explicitly,

$$\frac{P_n}{P_0} \leq \frac{((\epsilon + V)/\gamma)^n}{n!}, \quad (90)$$

and the sum of the expression on the right converges.

Does adding  $\epsilon$  guarantee a bimodal distribution? No. It removes the trap at zero, but the values of  $\epsilon$ ,  $V$ ,  $K$ ,  $h$ , and  $\gamma$  still determine the shape of the distribution. For suitable values, the cell can spend long periods at low expression and long periods at high expression, with stochastic transitions between them. The steady-state distribution can then have two peaks. Adding a small basal rate can also preserve two stable fixed points in the deterministic model, but their positions shift. The low fixed point is no longer exactly zero. A sufficiently large basal rate can remove the low state.

For the discrete model, we can locate the peaks by using Eq. 88. If  $P_{n+1}/P_n > 1$ , the probabilities rise as we go from  $n$  to  $n + 1$ . If the ratio is less than one, they fall. A peak occurs where the probabilities change from rising to falling (or at the boundary  $n = 0$  if  $P_1 < P_0$ ). We need two such peaks to call the distribution bimodal. Counting deterministic fixed points is not enough. Near zero copies, this exact discrete calculation is also safer than the continuous Fokker-Planck approximation.

Thus, we need basal production, or some other source of protein at zero, if this closed model is to keep switching between low and high expression in steady state. Without it, the system can still make several excursions before it reaches zero, but it cannot restart after reaching zero. We do not need basal production merely to obtain two stable deterministic fixed points. These are different questions.